

SNM INSTITUTE OF TECHNOLOGY

SEMINAR PRESENTATION

NEUROMORPHIC COMPUTING WITH SPIKING NEURAL NETWORKS: EFFICIENT DEPLOYMENT ON LOIHI 2 AND TRUENORTH

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Outline

1. Introduction
2. Motivation
3. Basic Terminologies
4. Taxonomy
5. Literature Review
6. Comparison Report
7. Conclusion
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- **Neuromorphic Computing** is a brain-inspired computing paradigm that mimics parallel, event-driven neural processing instead of traditional sequential logic.
- **Spiking Neural Networks (SNNs)** process information as sparse, discrete spikes in time, enabling event-driven, low-power computation.
- Neuromorphic chips such as **Intel's Loihi 2** and **IBM's TrueNorth** implement SNNs directly in hardware, exploiting temporal and spatial sparsity for efficiency.
- This seminar reviews four recent works on training, converting, and deploying SNNs on Loihi 2 and TrueNorth for speech, vision, sequence, and sensor-fusion tasks.

- Conventional CPU/GPU deep learning struggles with real-time, energy-constrained edge tasks such as autonomous driving, keyword spotting, and drone detection.
- Neuromorphic hardware keeps compute near memory and only communicates via sparse spikes, cutting redundant computation and data movement.
- Across the reviewed works, Loihi 2 and TrueNorth deliver order-of-magnitude gains in energy efficiency and latency over CPUs, GPUs, and edge accelerators such as NVIDIA Jetson.

- **Spiking Neural Network (SNN):** A network where neurons communicate via discrete, sparse spikes over time instead of continuous activations.
- **Loihi 2 / TrueNorth:** Neuromorphic processors with many small cores, each holding local neuron/synapse state, supporting asynchronous event-driven computation.
- **Sigma-Delta (Graded) Spikes:** Spikes that carry a quantized payload representing the *change* in a neuron's activation, exploiting temporal sparsity.
- **Synaptic Delay:** A learnable or fixed number of timesteps a spike takes to travel between neurons, enriching the temporal computation of an SNN.

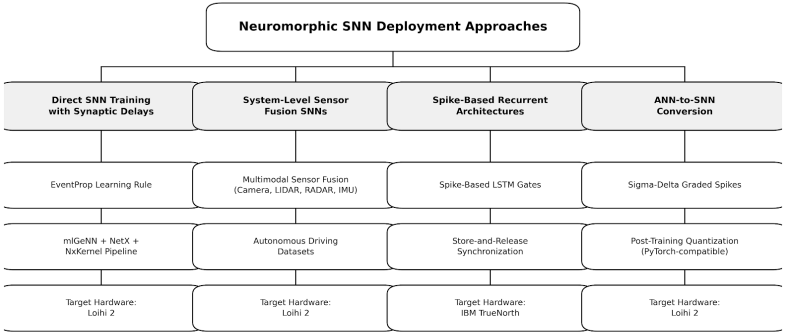


Figure: Taxonomy of Neuromorphic SNN Deployment Approaches Reviewed in this Seminar

1. A Complete Pipeline for Deploying SNNs with Synaptic Delays on Loihi 2

- Event-based training of SNNs with learnable synaptic delays using EventProp in the mlGeNN framework, exported via NetX for deployment on Loihi 2.
- Evaluates single-layer recurrent and 2-layer feedforward LIF-neuron SNNs, with and without delays.
- **Datasets:** Spiking Heidelberg Digits (SHD) and Spiking Speech Commands (SSC) keyword-spotting tasks.

1. Synaptic Delays on Loihi 2 — Key Results

- Delays give large feedforward accuracy gains (+13.3% SHD, +26.9% SSC); recurrent gains are more modest (+1.0%/+1.6%).
- 8-bit quantized Loihi 2 accuracy closely tracks the 32-bit mlGeNN GPU simulation.
- Up to 18× faster inference and roughly 250× lower energy-delay product than an NVIDIA Jetson Orin Nano.

1. Synaptic Delays on Loihi 2 (contd.)

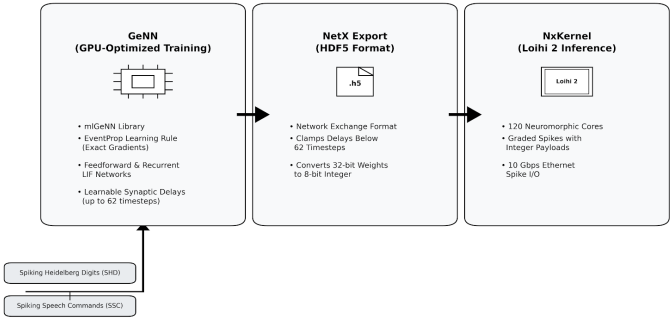


fig. 1.1 GeNN → NetX (HDF5) → NxKernel Pipeline for Training and Deploying Delay-SNNs on Loihi 2

2. Accelerating Sensor Fusion in Neuromorphic Computing: A Case Study on Loihi-2

- Implements five dataset-specific SNNs on Intel's Loihi-2 for multimodal sensor fusion in autonomous driving.
- **Approach:** Compares Loihi-2 against an Intel i9 12900H CPU and NVIDIA RTX 3060 GPU on throughput, power, and efficiency.
- **Datasets:** AIODrive, Oxford Radar RobotCar, D-Behavior (D-Set), nuScenes, Comma2k19.

2. Sensor Fusion on Loihi-2 — Key Results

- Loihi-2 reaches 103.94 GOP/s/W power efficiency vs. 0.30 (CPU) and 3.80 (GPU) GOP/s/W — over 100$\times$ and $\sim$30$\times$ higher.
- Inference times as low as 0.58–0.70 ms across datasets, beating prior FPGA/GPU implementations.
- Sustains sub-millisecond inference while drawing only 1.5–2.5 W across all five datasets.

2. Sensor Fusion on Loihi-2 (contd.)

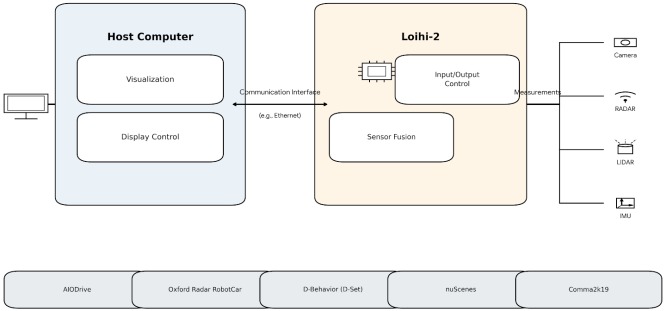


fig. 2.1 System Architecture Integrating the Loihi-2 Neuromorphic Chip for Advanced Sensor Fusion

3. A Spike-Based Long Short-Term Memory on a Neurosynaptic Processor

- Maps a standard LSTM (forget/input/output gates, cell state) onto IBM TrueNorth's spiking neurosynaptic cores.
- **Approach:** Weight discretization, piecewise-linear hard-sigmoid/hard-tanh gates, and a novel store-and-release synchronization mechanism.
- **Benchmarks:** Parity Check (XOR) and Embedded Reber Grammar (ERG) sequence tasks.

3. Spike-Based LSTM on TrueNorth — Key Results

- Accuracy rises with phase length (spike-rate precision), approaching the floating-point Keras/TensorFlow LSTM baseline of 100%.
- A 50-unit S-LSTM at 200-tick phase length consumes only 56 μJ/sample on TrueNorth.
- Up to 84–165$\times$ and 416–817$\times$ more energy-efficient than NVIDIA Tegra X1 and Titan X respectively.

3. Spike-Based LSTM on TrueNorth (contd.)

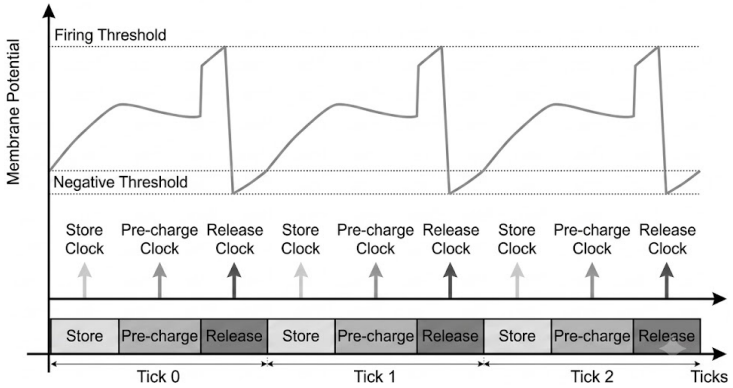


fig. 3.1 Store-and-Release Mechanism for Synchronizing the Spike-Based LSTM Gates

4. Sigma-Delta Neural Network Conversion on Loihi 2

- Converts trained, quantized ANNs into Sigma-Delta Neural Networks (SDNNs) using Loihi 2's graded spikes to exploit temporal and spatial sparsity.
- **Approach:** Redesigns the sigma-delta neuron microcode for fixed-point rescaling compatible with standard PyTorch quantization.
- **Applications:** Drone detection (YOLO-KP) on the Multi-Sensor Drone Detection dataset; steering-angle prediction (PilotNet) on the Driving dataset.

4. Sigma-Delta Conversion on Loihi 2 — Key Results

- Converted PilotNet SDNN achieves 16× lower energy and ∼40× lower energy-delay product than an ANN on Jetson Nano.
- Outperforms Intel's SLAYER-trained SDNN in mAP (0.62 vs. 0.37) while training about 11× faster per epoch.
- Field-deployed at US Navy Silent Swarm 2025: 112 FPS real-time inference at 5.0 ms latency and 20 mJ/frame.

4. Sigma-Delta Conversion on Loihi 2 (contd.)

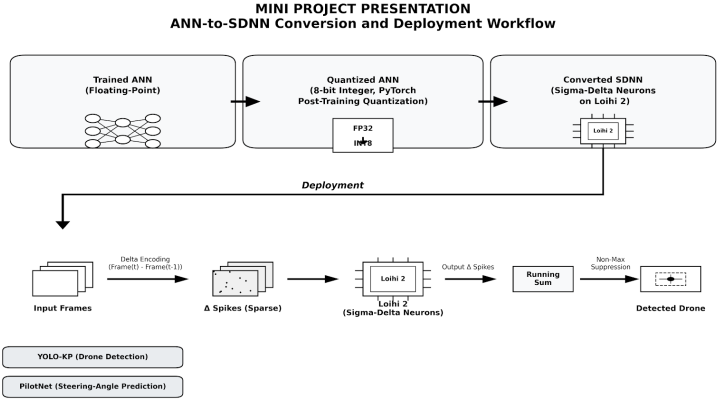


fig. 4.1 ANN-to-SDNN Conversion Workflow: Trained ANN → Quantized ANN → Converted SDNN on Loihi 2

Paper	Approach	Strength	Limitation
1	EventProp Delay Learning + mlGeNN/NetX	Large feedforward accuracy gains; up to 250× lower EDP	Delays capped at 62 timesteps
2	Five SNNs on Loihi-2 for Sensor Fusion	>100× CPU / ∼30× GPU energy efficiency	Tested on offline/synthetic data only
3	Spike-Based LSTM on TrueNorth	Up to 817× higher energy efficiency vs. Titan X	Higher per-sample latency than GPUs
4	ANN-to-SDNN Conversion via Graded Spikes	PyTorch-compatible; beats SLAYER training	Favors sparse, low-motion scenes

- Direct SNN training with delays, ANN-to-SNN conversion, spike-based recurrent architectures, and system-level sensor fusion all show that Loihi 2 and TrueNorth deliver large energy and latency gains over CPUs/GPUs.
- Each approach trades off differently: delays boost feedforward accuracy but are capped at 62 timesteps; sensor fusion is highly power-efficient but untested in real-time hardware; the S-LSTM's store-and-release scheme is efficient but latency-bound; SDNN conversion reuses standard training pipelines but favors sparse scenes.
- Future directions highlighted across these works: real-time hardware-in-the-loop sensor fusion, relaxing the synaptic-delay step limit, pairing SDNNs with event-based cameras, and reducing the precision/latency trade-off in spike-based recurrent models.

References I

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QUESTIONS?

Thank You
For your attention!